



# Luca Lazzaroni

Ricercatore a tempo determinato in Elettronica  
DITEN – UNIGE

SEDE      Genova, Italia

SETTORE    RTD-A · IINF-01/A

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## BIBLIOMETRIA

SCOPUS

|                        |                         |                       |                               |
|------------------------|-------------------------|-----------------------|-------------------------------|
| <b>65</b><br>DOCUMENTI | <b>329</b><br>CITAZIONI | <b>11</b><br>INDICE H | <b>18</b><br>ARTICOLI, 5 ANNI |
|------------------------|-------------------------|-----------------------|-------------------------------|

AGG. 21 SET 2026

### 01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università di Genova, settore IINF-01/A – Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni dal 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, LLM on-device. L'attività si svolge anche nell'ambito di progetti di ricerca europei e nazionali con partner industriali.

Edge AI

TinyML

Deep learning embedded

Guida automatizzata

Deep reinforcement learning

LLM on-device

### 02 Contatti e profili

|                                  |                                |                              |
|----------------------------------|--------------------------------|------------------------------|
| EMAIL<br>luca.lazzaroni@unige.it | UNIGE<br>rubrica.unige.it      | ORCID<br>0000-0001-8092-5473 |
| SCOPUS<br>57220892898            | GOOGLE SCHOLAR<br>w_6a6sYAAAAJ | GITHUB<br>@lucalazzaroni     |
| LINKEDIN<br>luca-lazzaroni       |                                |                              |

### 03 Incarichi e formazione

#### POSIZIONI

Mar 2025 – oggi

**Ricercatore a tempo determinato (RTD-A)**  
DITEN, Università di Genova  
Settore scientifico-disciplinare IINF-01/A — Elettronica.

|                       |                                                                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01/11/2023 – Feb 2025 | <b>Assegnista di ricerca</b><br>DITEN, Università di Genova<br>Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”. |
| Nov 2020 – 01/11/2023 | <b>Dottorando di ricerca</b><br>DITEN, Università di Genova – XXXVI ciclo<br>Borsa di dottorato: “Deep learning on the edge”.                                                      |

FORMAZIONE

|                                                                                                                                    |                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024                                                                                                                               | <b>Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni</b><br>Università di Genova – conseguito il 14 marzo 2024<br>Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning” |
| 2020                                                                                                                               | <b>Laurea Magistrale in Ingegneria Elettronica (LM-29)</b><br>Università di Genova – 24 luglio 2020<br>Tesi: “Progetto e implementazione di un motore di esecuzione multiplatforma per applicazioni IoT” · Voto: 110/110                                                                          |
| 2017                                                                                                                               | <b>Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)</b><br>Università di Genova – 27 ottobre 2017<br>Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”                                                                                           |
| 2014                                                                                                                               | <b>Diploma di Maturità Scientifica</b><br>Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)<br>Voto: 100/100                                                                                                                                                                           |
| Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020. |                                                                                                                                                                                                                                                                                                   |

04 Didattica e incarichi accademici

CORSI

|                        |                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------|
| A.A. 2024/25 – 2026/27 | <b>Distributed Electronic Systems and Technologies for Environmental Monitoring</b><br>Docente titolare · cod. 94631 |
| A.A. 2024/25 – 2026/27 | <b>Tiny Machine Learning</b><br>Co-titolare · cod. 114752                                                            |
| A.A. 2026/27           | <b>Dispositivi e Circuiti Elettronici</b><br>Docente titolare · cod. 65997                                           |
| dall'A.A. 2022/23      | <b>Machine Learning for Automated Driving</b><br>Cultore della materia · cod. 108716                                 |

|              |                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------|
| A.A. 2023/24 | <b>Computer Graphics Arch. — Games and Simulation</b><br>Docente a contratto · cod. 172317          |
| A.A. 2022/23 | <b>Autonomous Agents in Games</b><br>Supporto alla didattica · cod. 198216                          |
| A.A. 2020/21 | <b>Fondamenti di Programmazione di Sistemi Elettronici</b><br>Supporto alla didattica · cod. 166061 |
| A.A. 2020/21 | <b>Approccio Makers alla Progettazione Elettronica</b><br>Supporto alla didattica · cod. 186985     |

Tutti i corsi presso l'Università di Genova.

## SUPERVISIONE

- 1** Dottorando — Ali Zare, Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo)
- 12** Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)
- 10** Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)
- 7** Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)
- 2** Tesi di Laurea Magistrale — Informatica (LM-18)

## INCARICHI ACCADEMICI

**Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.**

## 05 Curatela e servizio

### RUOLI EDITORIALI

|      |                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Editor del volume</b><br>APPLEPIES 2026 — Springer LNEE<br>Atti in corso di pubblicazione.                                                                                 |
| 2025 | <b>Membro dell'Editorial Board (Early Career)</b><br>Sensors — MDPI<br>Dal 18 dicembre 2025.                                                                                  |
| 2025 | <b>Editor del volume</b><br>APPLEPIES 2025 — Springer LNEE<br>Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society. |

|      |                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | <b>Guest Editor</b><br>Energies — MDPI<br>Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025. |
|------|-------------------------------------------------------------------------------------------------------------------------------------|

ORGANIZZAZIONE DI CONVEGNI

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| <b>Program Chair</b><br>APPLEPIES 2026<br>Bologna, Italia, 10–11 settembre 2026.                                |
| <b>Program Chair</b><br>APPLEPIES 2025<br>Torino, Italia, 11–12 settembre 2025.                                 |
| <b>Program Committee</b><br>APPLEPIES<br>Dal 2023.                                                              |
| <b>Program Committee</b><br>GALA — Games and Learning Alliance<br>Dal 2023.                                     |
| <b>Chairman di sessione</b><br>APPLEPIES 2023<br>1st Short Oral session — Genova, Italia, 28–29 settembre 2023. |

ATTIVITÀ DI REVISIONE

| RIVISTE                                                      | CONFERENZE                                                      |
|--------------------------------------------------------------|-----------------------------------------------------------------|
| IEEE Access                                                  | IEEE ICRA — International Conference on Robotics and Automation |
| Journal of Systems Architecture                              | IEEE/RSJ IROS — Intelligent Robots and Systems                  |
| Engineering Applications of Artificial Intelligence          | GALA — Games and Learning Alliance                              |
| Sensors                                                      | APPLEPIES                                                       |
| IET Intelligent Transport Systems                            |                                                                 |
| IEEE Transactions on Intelligent Vehicles                    |                                                                 |
| IEEE Transactions on Circuits and Systems II: Express Briefs |                                                                 |
| Robotics and Autonomous Systems                              |                                                                 |
| Scientific Reports                                           |                                                                 |
| Journal of Big Data                                          |                                                                 |

**06 Interventi a convegni**

13 INTERVENTI

|                |                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------|
| 11-12 set 2025 | <b>APPLEPIES 2025</b><br>Torino, Italia                                                            |
| 11-14 gen 2025 | <b>43rd IEEE ICCE — International Conference on Consumer Electronics</b><br>Las Vegas, NV, USA     |
| 19-20 set 2024 | <b>APPLEPIES 2024</b><br>Torino, Italia                                                            |
| 24-26 giu 2024 | <b>55th Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Genova, Italia           |
| 28-29 set 2023 | <b>APPLEPIES 2023</b><br>Genova, Italia                                                            |
| 7-9 set 2023   | <b>54th Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Noto (RG), Italia        |
| 26-27 set 2022 | <b>APPLEPIES 2022</b><br>Genova, Italia                                                            |
| 7-9 set 2022   | <b>53rd Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Pizzo (VV), Italia       |
| 21-22 set 2021 | <b>APPLEPIES 2021</b><br>Pisa, Italia                                                              |
| 1-2 dic 2021   | <b>GALA 2021 — Games and Learning Alliance</b><br>La Spezia, Italia (online)                       |
| 7-9 lug 2021   | <b>52nd Annual Meeting of the Italian Society of Electronics (SIE)</b><br>Trieste, Italia (online) |
| 9-10 dic 2020  | <b>GALA 2020 — Games and Learning Alliance</b><br>Laval, Francia (online)                          |
| 19-20 nov 2020 | <b>APPLEPIES 2020</b><br>Online                                                                    |

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## 07 Premi e associazioni

2025

### **Best Presentation of the Session — sessione “Edge Computing”**

43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

## **ASSOCIAZIONI PROFESSIONALI**

### **IEEE**

IEEE Member e IEEE Young Professional.

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### **IEEE Consumer Technology Society**

Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

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### **SIE — Società Italiana di Elettronica**

Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

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## 08 Progetti di ricerca

Synergies

### **Generation of scenarios for development, training, virtual testing and validation of CCAM systems**

Europeo — HORIZON-CL5-2023-D6-01-02 · Task Leader, WP8 — Evaluation Framework (T8.1)

Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore.

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Hi-Drive

### **Addressing challenges toward the deployment of higher automation**

Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova

Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.

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L3Pilot

### **Piloting Automated Driving on European Roads**

Europeo — H2020 · Membro dell'unità di ricerca dell'Università di Genova

Sperimentazione su larga scala di funzioni di guida automatizzata SAE L3 su strade pubbliche europee, con le principali case automobilistiche e fornitori europei.

**Green Machine Learning for the IoT**

Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)

Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.

**Assistente Digitale Integrato Automotive**

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

2026

**A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction**

L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti

*IEEE Signal Processing Letters* vol. 33, pp. 136–140

Rivista

**FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction**

A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srour, F. Tango, F. Bellotti

*IEEE Transactions on Circuits and Systems II: Express Briefs* vol. 73, no. 8, pp. 888–892

Rivista

**Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms**

H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta

*IEEE Access* vol. 14, pp. 44886–44903

Rivista

**StabilEdge, Formal Verification of Edge AI Applications**

**L. Lazzaroni**, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti

*IEEE Access* vol. 14, pp. 37416–37434

Rivista

**Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers**

A. Saad, F. Bellotti, H. Ballout, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering  
pp. 321–327

Conferenza

**Preface**

M. Ruó Roch, R. Berta, A. Gómez, **L. Lazzaroni**, M. Martina, P.M. Ros, N. Pazos Escudero

*Lecture Notes in Electrical Engineering* pp. v–vi

Editoriale

**Optimizing Binary AutoEncoders on STM32 Microcontrollers**

H. Ballout, R. Berta, **L. Lazzaroni**, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering  
pp. 301–307

Conferenza

**Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students**

L. Motta, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, N. Busi

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering  
pp. 242–248

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### Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, **L. Lazzaroni**, M. Fresta, L. Forneris, A. Pighetti, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 218–223

Conferenza

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### Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, **L. Lazzaroni**, H. Ballout, O. Srour, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 25–31

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### Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, **L. Lazzaroni**, A. Pilla, H. Rojhan, L. Forneris

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 212–217

Conferenza

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### Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, **L. Lazzaroni**, M. Fresta, A. Saad, F. Bellotti

*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 289–294

Conferenza

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### FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, **L. Lazzaroni**, O. Srour, R. Berta

*Proceedings of SIE 2025* Lecture Notes in Electrical Engineering pp. 295–301

Conferenza

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### Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 85–90

Conferenza

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### Embedded Deployment of an LLM-Based Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. Pighetti, A. Saad, O. Srour, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 273–278

Conferenza

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### Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 286–291

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### A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 183–190

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### A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 249–254

Conferenza

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### A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 79–84

Conferenza

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**An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection**

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

*IEEE Sensors Journal* vol. 25, no. 24, pp. 44859–44874

Rivista

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**TinyML Acceleration with MAX78000**

A. Dabbous, L. Lazzaroni, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

*Proceedings of SIE 2024* Lecture Notes in Electrical Engineering pp. 468–474

Conferenza

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**Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection**

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, L. Lazzaroni

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 156–163

Conferenza

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**Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking**

H. Ballout, R. Berta, I. Bochenko, L. Lazzaroni, A. Dhonthi, V. Hashemi, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 387–394

Conferenza

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**Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking**

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, L. Lazzaroni

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 139–147

Conferenza

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**RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset**

R. Berta, F. Bellotti, M. Cossu, L. Forneris, L. Lazzaroni, A. Pighetti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 395–402

Conferenza

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**RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking**

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, L. Lazzaroni, C. Oh

*IEEE Sensors Letters* vol. 9, no. 9, art. 6010204

Rivista

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**Quantization of MobileNetV2 for Resource-Constrained Microcontrollers**

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 497–504

Conferenza

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**Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board**

H. Ballout, R. Berta, A. Dabbous, M. Fresta, L. Lazzaroni, A. Pighetti, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 242–249

Conferenza

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**On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection**

M. Fresta, F. Bellotti, L. Lazzaroni, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

*IEEE Open Journal of the Industrial Electronics Society* vol. 6, pp. 1351–1369

Rivista

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**Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices**

A. Capello, R. Berta, M. Fresta, L. Lazzaroni, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 323–330

Conferenza

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### Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 339–346

Conferenza

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### Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, L. Lazzaroni, C. Oh

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 123–130

Conferenza

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### Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, L. Lazzaroni, G. Merthiot, F. Tango, R. Berta

*IEEE Access* vol. 13, pp. 26589–26607

Rivista

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### Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, L. Lazzaroni, J.L. Sauvaget, H. Touil, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 250–257

Conferenza

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### Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti

*2025 IEEE International Conference on Consumer Electronics (ICCE)*

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conferenza

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### Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, L. Lazzaroni, A. Pighetti, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 234–241

Conferenza

2024

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### An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

*Engineering Applications of Artificial Intelligence* vol. 136, art. 108998

Rivista

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### Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, L. Lazzaroni, C. Oh

*ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving*

Workshop

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### CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, L. Lazzaroni, F. Bellotti

*Electronics* vol. 13, no. 9, art. 1624

Rivista

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### End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, L. Lazzaroni, F. Ansovini, R. Berta

*Electronics* vol. 13, no. 7, art. 1286

Rivista

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### YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, L. Lazzaroni, J.L. Sauvaget, F. Bellotti

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 418–423

Conferenza

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### Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 444–449

Conferenza

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### Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

*Applications in Electronics Pervading Industry, Environment and Society* Lecture Notes in Electrical Engineering

pp. 124–129

Conferenza

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### Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

*Journal of Intelligent and Connected Vehicles* vol. 7, no. 3, pp. 229–244

Rivista

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### Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 946–960

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### Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

**L. Lazzaroni**, A. Pighetti, F. Bellotti, R. Berta

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### Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

*IEEE Open Journal of the Industrial Electronics Society* vol. 5, pp. 781–794

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### Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

**L. Lazzaroni**, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

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### Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

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### Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

*International Journal of Serious Games* vol. 10, no. 4, pp. 153–170

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### Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

*Applied Sciences* vol. 13, no. 11, art. 6432

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### Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

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### Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

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### Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

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### Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

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### Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

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### Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

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### Deep Reinforcement Learning for Automated Car Parking

**L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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### Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

*53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)*

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### Investigating High-Level Decision Making for Automated Driving

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### Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

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### An Embedded End-to-End Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta

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### Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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### High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

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### Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

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### Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta

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### A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

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### Project and Development of an Italian Offline Embedded Voice Assistant

**L. Lazzaroni**, F. Bellotti, A. De Gloria, R. Berta

*52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste*

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### Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

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### Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

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### Employing an IoT Framework as a Generic Serious Games Analytics Engine

**L. Lazzaroni**, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

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